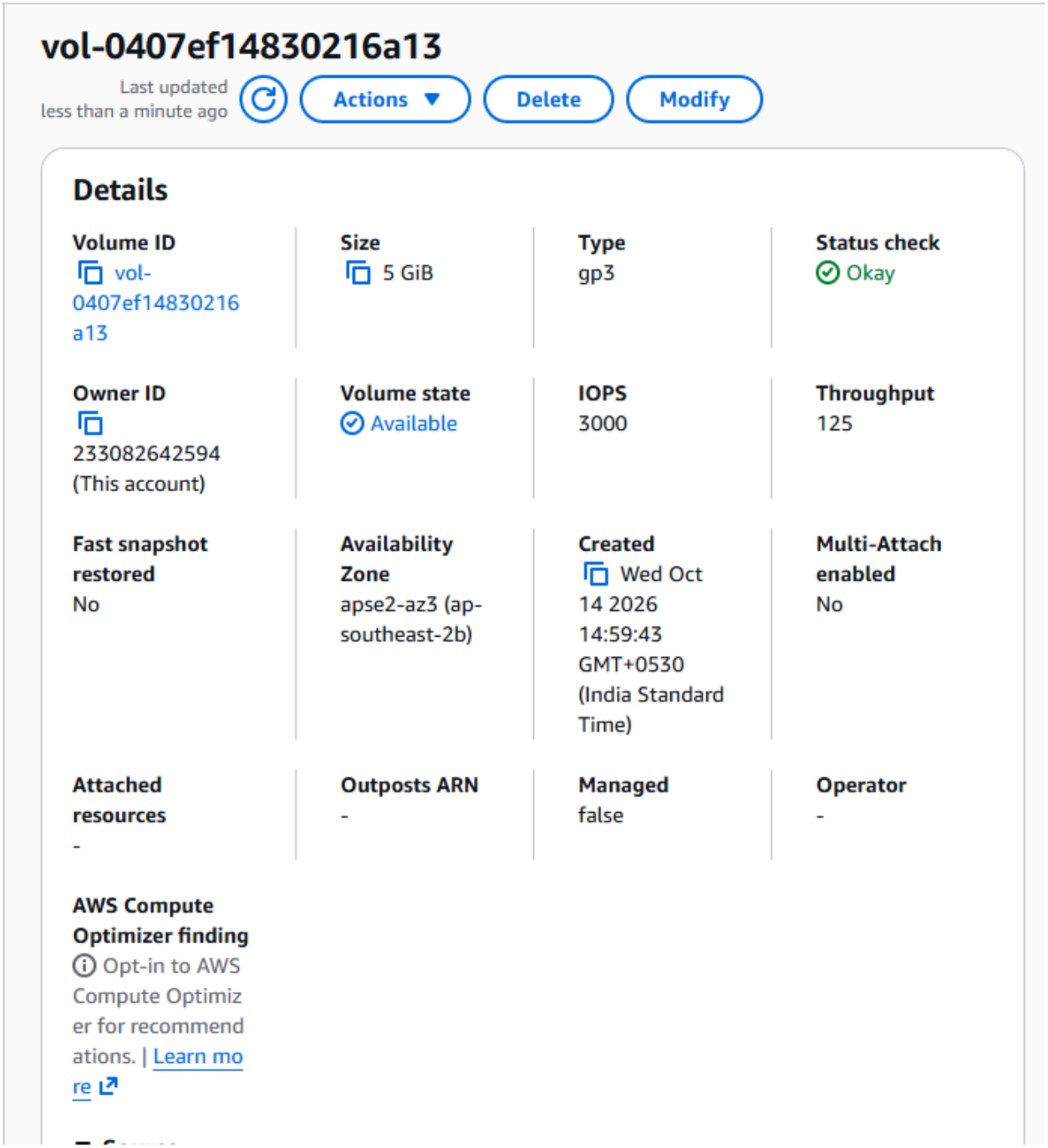
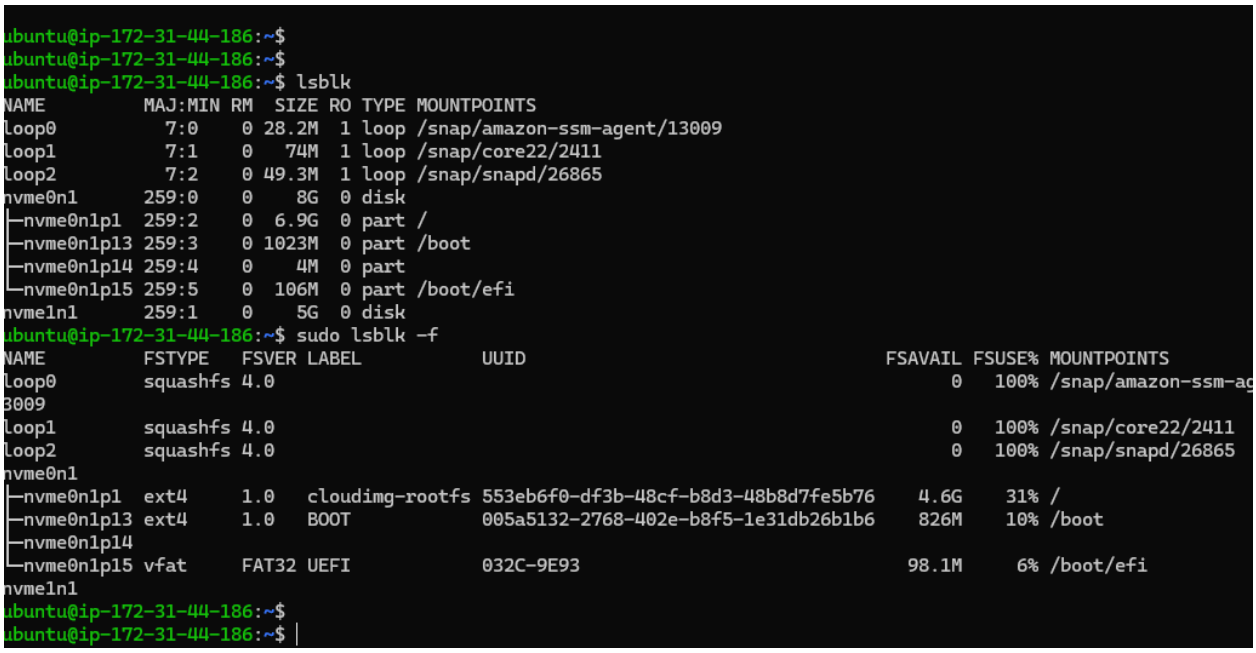
AWS EBS Snapshot lab
Hrishinandan p

Tasks

1. Create a t2.micro Ubuntu EC2 instance. & Create a 5 GB gp3 EBS volume in the same AZ.

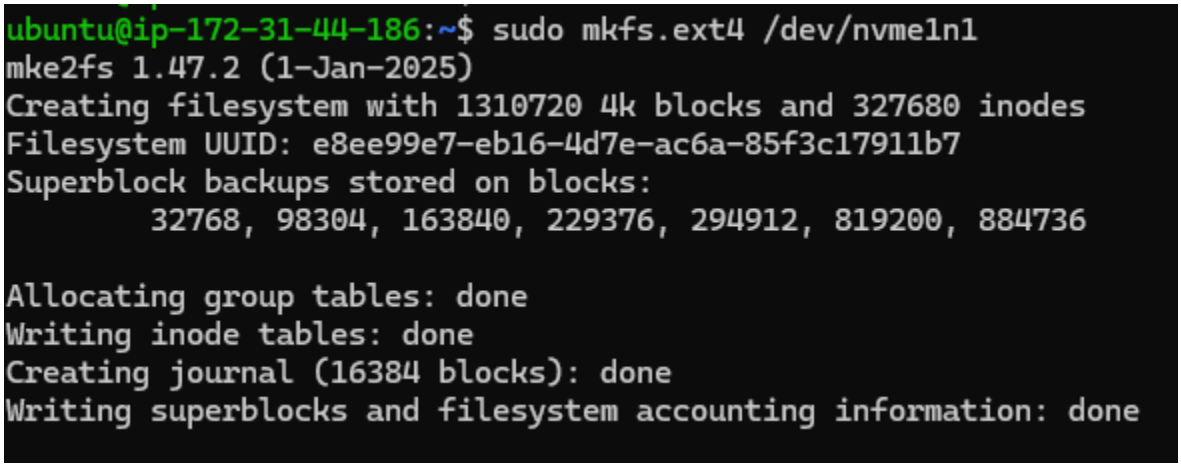


Attach it as /dev/xvdf. Verify using lsblk.,



```
ubuntu@ip-172-31-44-186:~$
ubuntu@ip-172-31-44-186:~$
ubuntu@ip-172-31-44-186:~$ lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0          7:0    0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1          7:1    0   74M  1 loop /snap/core22/2411
loop2          7:2    0 49.3M  1 loop /snap/snapd/26865
nvme0n1      259:0    0    8G  0 disk
├─nvme0n1p1  259:2    0  6.9G  0 part /
├─nvme0n1p13 259:3    0 1023M  0 part /boot
├─nvme0n1p14 259:4    0    4M  0 part
└─nvme0n1p15 259:5    0  106M  0 part /boot/efi
nvme1n1      259:1    0    5G  0 disk
ubuntu@ip-172-31-44-186:~$ sudo lsblk -f
NAME         FSTYPE   FSVER LABEL           UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0        squashfs 4.0                                                           0   100% /snap/amazon-ssm-ag
3009
loop1        squashfs 4.0                                                           0   100% /snap/core22/2411
loop2        squashfs 4.0                                                           0   100% /snap/snapd/26865
nvme0n1
├─nvme0n1p1  ext4     1.0   cloudimg-rootfs 553eb6f0-df3b-48cf-b8d3-48b8d7fe5b76    4.6G    31% /
├─nvme0n1p13 ext4     1.0   BOOT            005a5132-2768-402e-b8f5-1e31db26b1b6    826M    10% /boot
├─nvme0n1p14
└─nvme0n1p15 vfat     FAT32 UEFI            032C-9E93                              98.1M     6% /boot/efi
nvme1n1
ubuntu@ip-172-31-44-186:~$
ubuntu@ip-172-31-44-186:~$
```

Format: sudo mkfs.ext4 /dev/xvdf.



```
ubuntu@ip-172-31-44-186:~$ sudo mkfs.ext4 /dev/nvme1n1
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: e8ee99e7-eb16-4d7e-ac6a-85f3c17911b7
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

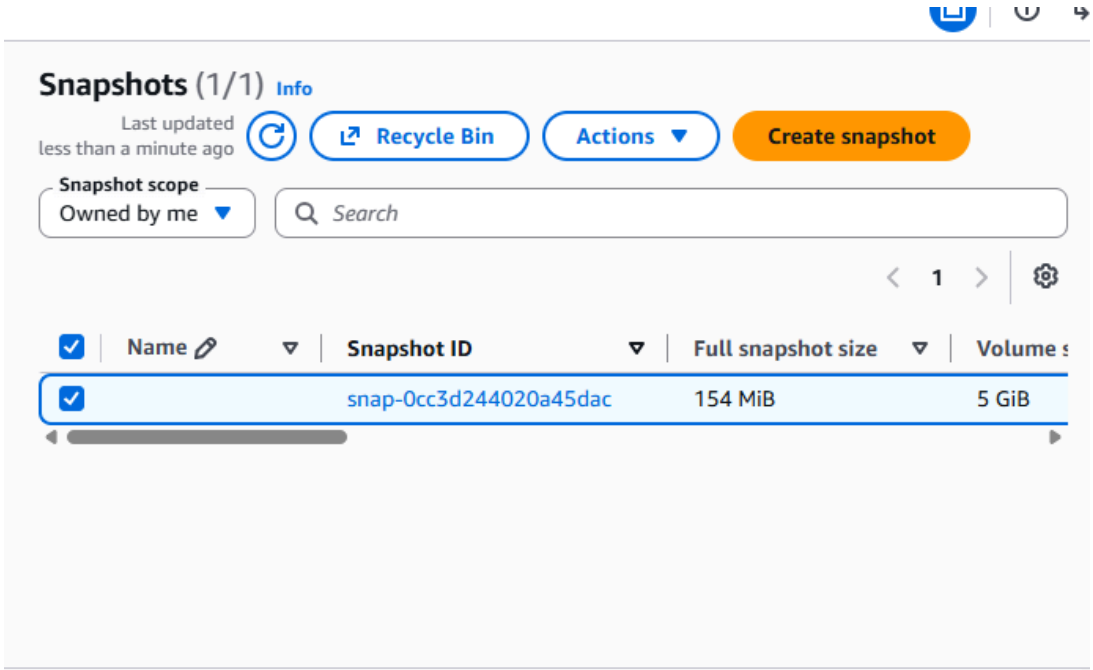
Mount at /data and verify with df -h.

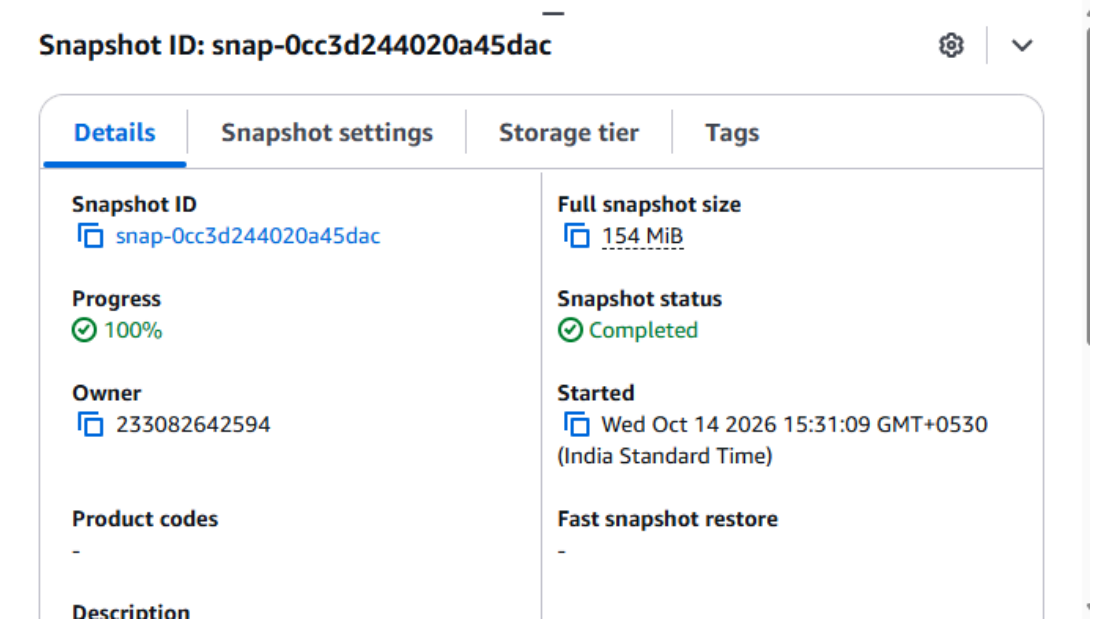
```
ubuntu@ip-172-31-44-186:~$ sudo mkdir -p /data
ubuntu@ip-172-31-44-186:~$ sudo mount /dev/nvme1n1 /data
ubuntu@ip-172-31-44-186:~$ df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.1G  4.6G  31% /
tmpfs            455M     0  455M   0% /dev/shm
tmpfs            182M  900K  181M   1% /run
efivarfs         128K  3.4K  120K   3% /sys/firmware/efi/efivars
tmpfs            455M     0  455M   0% /tmp
none             1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none             1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M   96M  827M  11% /boot
/dev/nvme0n1p15  105M  6.3M   99M   7% /boot/efi
none             1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
none             1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
none             1.0M     0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs             91M  8.0K   91M   1% /run/user/1000
/dev/nvme1n1     4.9G  1.3M  4.6G   1% /data
ubuntu@ip-172-31-44-186:~$
```

Create sample files under /data.

```
ubuntu@ip-172-31-44-186:~$ echo "Hello AWS EBS Snapshot Lab" | sudo tee /data/file1.txt
Hello AWS EBS Snapshot Lab
ubuntu@ip-172-31-44-186:~$ echo "This is my backup test" | sudo tee /data/file2.txt
This is my backup test
ubuntu@ip-172-31-44-186:~$ ls -l /data
total 24
-rw-r--r-- 1 root root    27 Oct 14 09:47 file1.txt
-rw-r--r-- 1 root root    23 Oct 14 09:49 file2.txt
drwx------ 2 root root 16384 Oct 14 09:43 lost+found
ubuntu@ip-172-31-44-186:~$ cat /data/file1.txt
Hello AWS EBS Snapshot Lab
ubuntu@ip-172-31-44-186:~$ cat /data/file2.txt
This is my backup test
ubuntu@ip-172-31-44-186:~$
```

Create an EBS snapshot named snapshot-lab-backup

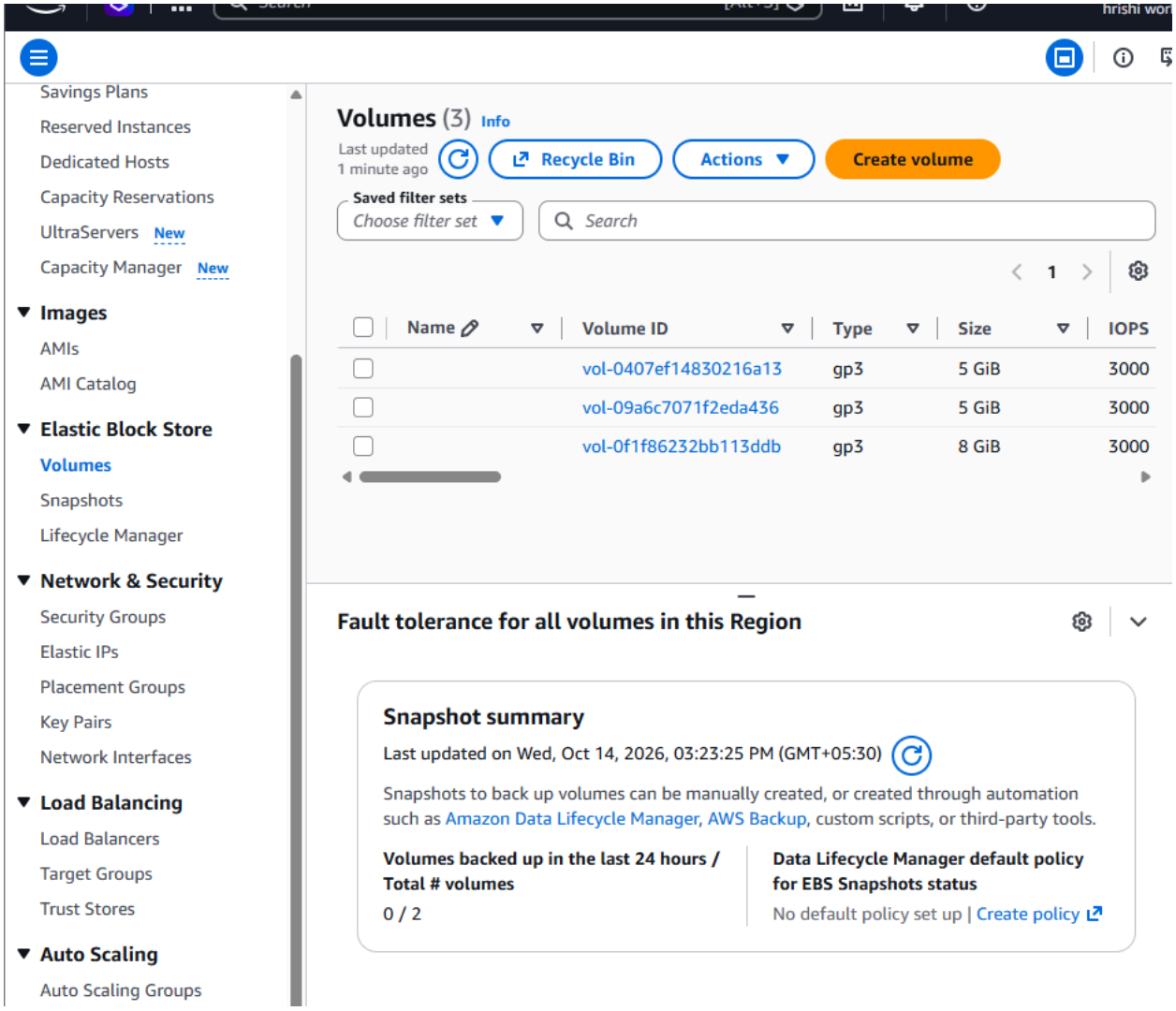




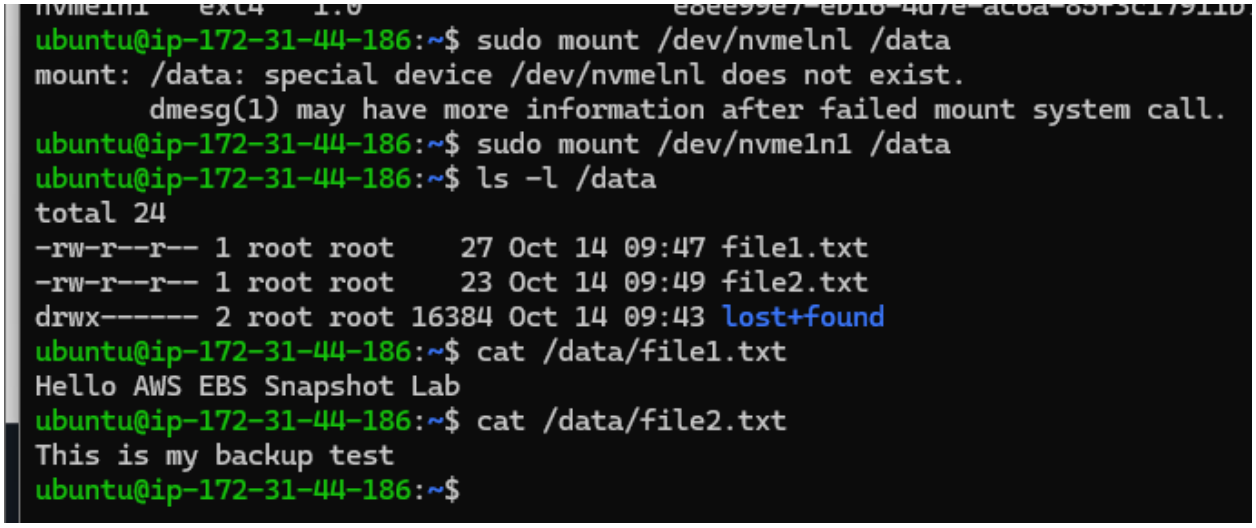
Delete the files to simulate data loss.

```
ubuntu@ip-172-31-44-186:~$ sudo rm -f /data/file1.txt /data/file2.txt
ubuntu@ip-172-31-44-186:~$ ls -l /data
total 16
drwx------ 2 root root 16384 Oct 14 09:43 lost+found
ubuntu@ip-172-31-44-186:~$
```

Create a new volume from the snapshot and Detach old one.



Mount it again and verify the files are restored.



What is an EBS snapshot?

A point-in-time backup of an EBS volume.

Are snapshots incremental?

Yes, subsequent snapshots save changed and new data blocks.

Can you restore to a larger volume?

Yes, but the volume must be at least as large as the snapshot's original volume.

Does deleting a volume delete its snapshots?

No, snapshots are independent of the volume.

How do snapshots help disaster recovery?

They let us create replacement volumes and recover data from an earlier point in time.